

SOC INCIDENT INVESTIGATION REPORT

Incident 4 — Multi-stage incident on one endpoint

Controlled adversary simulation / Microsoft Defender for Endpoint / Atomic Red Team

1. Executive Summary

On 14 August 2026, a controlled adversary simulation was executed on endpoint user1 using Atomic Red Team technique T1059.001-17 (PowerShell Command Execution). The test launched PowerShell through cmd.exe using an encoded command. Microsoft Defender for Endpoint collected process and command-line telemetry and correlated the activity into Incident 4, generating high- and medium-severity detections.

The activity was authorized laboratory activity rather than an actual compromise. The investigation nevertheless demonstrates a realistic SOC workflow: identify the initiating account and process chain, reconstruct execution, validate it against known test context, assess impact, and translate the observed behavior into a detection hypothesis.

No destructive action was observed from this test, and the endpoint remained available for continued controlled testing.

2. Case Details

Incident ID	4
Endpoint	user1
Account	user1\thato
Primary technique	T1059.001 — Command and Scripting Interpreter: PowerShell
Atomic test	T1059.001-17 — PowerShell Command Execution
Incident severity	High
Alerts	Potential human-operated malicious activity; Suspicious behavior by cmd.exe was observed
Detection source	Microsoft Defender for Endpoint / EDR
Simulation source	Atomic Red Team
Activity date	14 August 2026

3. Investigation Objective

- Determine what executed, under which account, and through which process chain.
- Correlate Atomic Red Team execution with Microsoft Defender telemetry.
- Determine whether the observed activity represented a genuine compromise or an authorized simulation.

- Identify behavioral indicators suitable for a detection rule.
- Create a baseline for later detection tuning and malware investigation.

4. Timeline and Evidence

Time	Source	Evidence	Assessment
13:31:10 UTC	Atomic Red Team	T1059.001-17 executed successfully; benign output; exit code 0.	Test execution confirmed.
15:31:09 Defender time	MDE telemetry	cmd.exe launched powershell.exe with an encoded -e command under user1\thato; PID 7856.	Strong process-chain evidence.
15:33:23 Defender time	MDE telemetry	SenseIR.exe executed a Defender data-collection PowerShell script.	EDR collection activity; not itself evidence of compromise.
14 Aug 2026	MDE Incident 4	Incident created at 15:31:37; last activity 15:42:26.	Alerts correlated into one endpoint incident.

Time note: the VM clock and analyst local time differed by approximately two hours. Use Defender timestamps as the primary correlation reference and document the time-zone context in screenshots.

5. Process Execution Analysis

Observed process chain: cmd.exe → powershell.exe -e <encoded command>

The encoded PowerShell command was visible in Defender telemetry. PowerShell executed successfully and produced the expected benign test output. Execution occurred under user1\thato, matching the account used for the authorized test.

Encoded PowerShell is not, by itself, proof of malicious activity. Confidence increases when combined with suspicious parent processes, unusual users, downloads, persistence, credential access, network activity, or other malicious behavior.

6. Defender Detection and Correlation

Microsoft Defender for Endpoint generated and correlated two alerts:

- Potential human-operated malicious activity — High.
- Suspicious behavior by cmd.exe was observed — Medium.

The alerts were correlated into Incident 4: Multi-stage incident on one endpoint. This allowed investigation from alert to account, endpoint, process tree, command line, and related activity.

7. Findings

Finding	Status	Assessment
Execution	Confirmed	Atomic test completed successfully with exit code 0.
Encoded PowerShell	Confirmed	Defender captured powershell.exe using the -e encoded-command parameter.
Parent process	Confirmed	cmd.exe launched PowerShell.
Account attribution	Confirmed	Activity was associated with user1\thato.
EDR visibility	Confirmed	Defender collected process/command-line telemetry and generated correlated alerts.
Real compromise	Not supported	The activity was a known, authorized adversary simulation.
Destructive impact	None observed	The test produced benign output and did not intentionally damage the VM.

8. Initial Detection Engineering Hypothesis

Behavior: PowerShell + encoded command + suspicious parent process

The analytic should target the behavior rather than the Atomic test number or exact encoded string. The initial hypothesis is to identify PowerShell executions containing encoded-command indicators such as -enc, -encodedcommand, or -e, then increase confidence when the parent process or surrounding process tree is unusual.

9. Baseline Microsoft Defender Advanced Hunting Query

```
DeviceProcessEvents
| where Timestamp > ago(24h)
| where FileName =~ "powershell.exe"
| where ProcessCommandLine has_any ("-enc", "-encodedcommand", "-e ")
| project Timestamp, DeviceName, AccountName, InitiatingProcessFileName,
    InitiatingProcessCommandLine, FileName, ProcessCommandLine,
    ProcessId, InitiatingProcessId
| order by Timestamp desc
```

This is a baseline analytic, not a production-ready rule. It is expected to produce false positives because encoded PowerShell can be legitimate. Subsequent tests should be used to measure false positives and add contextual conditions.

10. Detection Tuning Plan

- Baseline normal encoded PowerShell usage.
- Increase confidence when the parent process is unusual or unrelated to normal administration.
- Correlate with downloads, network connections, persistence, credential-access behavior, or suspicious child processes.

- Preserve device and account context for immediate analyst triage.
- Avoid suppressing an entire account or endpoint simply because it is used for testing.
- Record true positives and false positives from subsequent tests and tune iteratively.

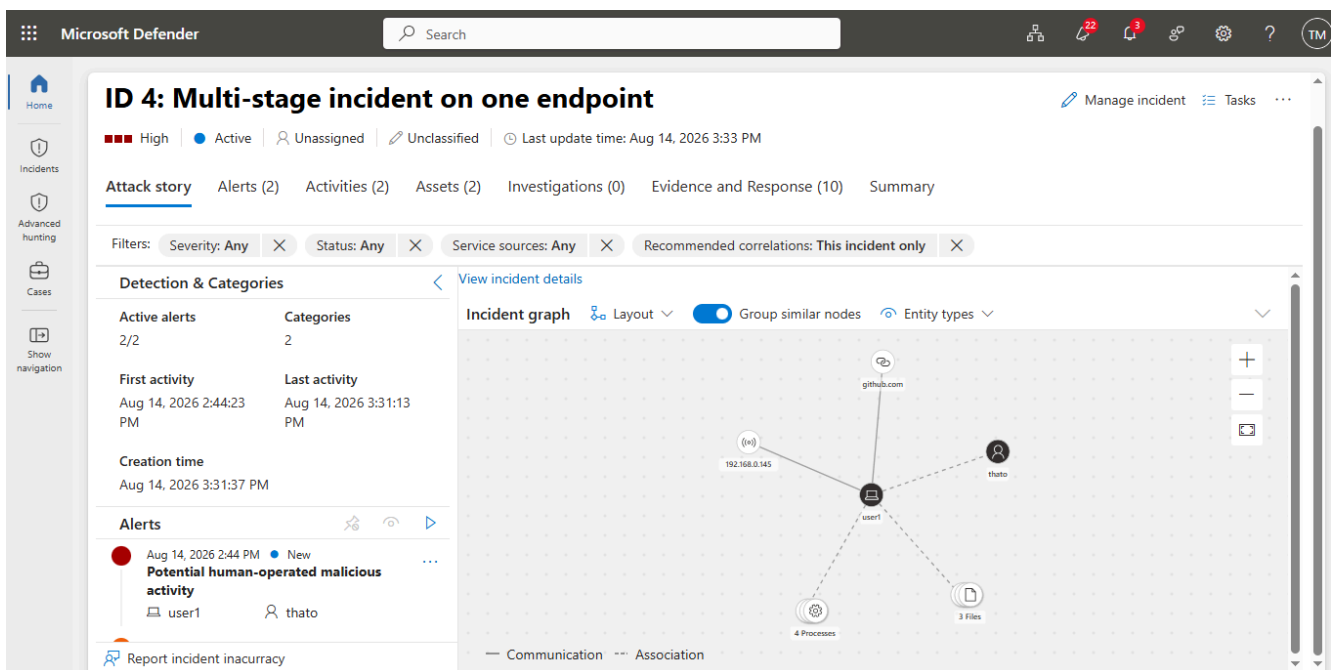
11. Analyst Assessment and Disposition

Assessment: TRUE POSITIVE for the behavioral detection, but NOT evidence of real-world compromise. The execution was an authorized Atomic Red Team simulation.

Recommended disposition: close the individual incident after preserving evidence and documenting the test identifier. Retain the detection as a candidate analytic for further validation rather than treating this single execution as sufficient for production deployment.

12. Screenshot / Evidence Placeholders

Screenshot A — Incident 4 overview and severity.



Analyst notes:

Screenshot B — Potential human-operated malicious activity alert. Suspicious behavior by cmd.exe alert.

+ Create

↓ Export

📅 6 Months

🔍 Report incident inaccuracy

3 Alerts

🔍 Search for name or ID

⚙️ Customize columns

Filter set:

🔍 Add filter

☐	Alert name	Tags	Severity	Investigation state	Status	Category	Impacted assets	Correlation reasc
☐	A file or network connection related to a ra...	Ransomware	High		Resolved	Ransomware	user1	Same device (2 aler...
☐	Potential human-operated malicious activity		High	Remediated	Resolved	Suspicious activity	user1	thato Same device (2 aler...
☐	Suspicious behavior by cmd.exe was observ...		Medium		Resolved	Command and Con...	user1	thato Same device (2 aler...

Analyst notes:

13. Limitations

- Only one Atomic test has been evaluated at this stage.
- The activity was controlled and therefore does not establish performance against genuine malware.
- The baseline detection has not yet been tested for false-positive rate.
- The VM clock differed from analyst local time; Defender portal timestamps should be used for final correlation.
- Credential-access, persistence, lateral-movement, and malware detections remain unvalidated until later test phases.

14. Case Conclusion

Incident 4 demonstrates that Microsoft Defender for Endpoint provided sufficient telemetry to reconstruct a controlled encoded-PowerShell execution and correlate it into a multi-stage incident. The investigation supports a defensible behavioral detection hypothesis centered on encoded PowerShell combined with suspicious process context. The individual case is sufficiently investigated for documentation and controlled closure, while the analytic should remain under validation as additional test cases are introduced.